Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/05/2008
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Effect of Spaceflight on Microbial Gene Expression and Virulence		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Cheryl.Nickerson@asu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	480-727-7520
Organization Name:	Arizona State University		
PI Address 1:	Center for Infectious Diseases and Vaccinology/The Biodesign Institute		
PI Address 2:	1001 S McAllister Avenue		
PI Web Page:	https://		
City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	06/01/2006	**End Date:**	01/31/2009
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	NOTE: End date changed to 1/31/2009 per A. Chu/ARC (6/08)		
Key Personnel Changes/Previous PI:	PI moved to Arizona State University from Tulane University mid-2006.		
COI Name (Institution):	Schurr, Michael (University of Colorado Health Sciences Center) Buchanan, Kent (Oklahoma City University)		
Grant/Contract No.:	NNA06CB26A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Human presence in space, whether permanent or transient, will be accompanied by the presence of microbes. The changes that occur to microorganisms during space flight are not fully understood, and the potential of a negative impact upon health, safety, and performance of crewmembers requires further investigation. Space flight studies have demonstrated changes in bacterial and fungal growth, morphology, and metabolism, and ground-based studies have indicated changes in bacterial virulence characteristics and protein production during growth in modeled microgravity, a laboratory environment that stimulates some aspects of spaceflight. Thus, during long-duration space flight, the virulence of pathogenic microbes may be enhanced and normally harmless commensal or environmental microbes may become pathogenic. This effect would be intensified by several factors, including: 1) the increased duration of missions, 2) a greater microbial diversity resulting from the use of regenerative life support systems, and 3) a potentially compromised human immune system during flight. Investigation into the effects of space flight on microbial characteristics is difficult as flight experiments are expensive and opportunities are rare. Therefore, ground-based analog systems have been developed for cellular and microbial testing; the most popular system being NASA designed rotating wall vessel (RWV) bioreactors, which simulate many of the mechanistic effects of microgravity on biological systems. The objective of this research is to compare microbial gene expression and virulence attributes in microorganisms during extended space flight to the same organisms cultured under normal gravity and in the modeled microgravity of the RWV. The latter comparison is essential to verify the applicability of the RWV bioreactors to simulate in-flight conditions and to use this system to access the probability of in-flight infectious disease. This proposal is to fly two copies of the automated single locker MOBIAS (Multiple Orbital Bioreactor with Instrumentation and Automated Sampling) culture unit, to allow simultaneous analysis in multiple replicates of four difference bacterial and fungi. This study develops a foundation from which future in-flight research can be based and which may also hold commercial promise for translation to the clinical bedside. The objective of this research is to determine changes in microbial gene expression and virulence attributes during extended spaceflight. These results will be compared to those obtained when the same experiments are performed with the identical microbes cultured under ground-based conditions of modeled microgravity (MMG) in NASA-designed bioreactors. This comparison is vital in determining the value of the NASA-designed bioreactors in suspension culture of microbes under MMG for use in: (a) simulating the probability of in-flight infectious disease; and, (b) developing commercial potential for the bioreactors for translation to the clinical bedside. Four model microbes, two bacterial and two fungal, are proposed for use in these studies: a). the obligate pathogen Salmonella typhimurium, b). the opportunistic pathogen, Pseudomonas aeruginosa, c). the commensal organism and opportunistic pathogen, Candida albicans; and, d). the classic model eukaryotic microbe, Saccharomyces cerevisiae. These organisms were shown to be gravity sensitive in earlier flight experiments and/or ground-based simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will develop a foundation from which future ground-based research can be based and also holds commercial promise for translation to the clinical bedside. These investigations will be the first of their kind to investigate comprehensive changes in microbial gene expression which may occur during extended spaceflight and their comparison to the expression profiles of the same microbes grown as ground-based suspension cultures under MMG in the NASA-designed bioreactors. The latter comparison is essential to verify the applicability of the MMG bioreactors to simulate in-flight conditions and to use this system to assess the probability of in-flight infectious disease. However, these studies will also provide a solid foundation for testing the commercial applicability of ground-based microbial suspension culture under MMG for the development of vaccines and other novel countermeasures, which are not achievable by any other ground-based means, for the treatment and prevention of infectious diseases occurring on Earth. For example, the PI has previously demonstrated that when cultivated under MMG, the enteric pathogen, S. typhimurium, is more potent in terms of its disease-causing potential. Expanding this work to include other important pathogens could provide significant insight about the microbial response to MMG, and could provide clues for the functioning of known virulence systems, or the identification of novel, uncharacterized microbial virulence strategies, which could lead directly to the development of countermeasure for infectious human disease. Importantly, the knowledge gained from these flight and ground-based studies has identified the existence of a previously unknown mechanism that affects microbial virulence. By targeting and understanding this new mechanism, novel therapeutics and vaccines may be developed against infectious disease.
Task Progress:	Assessing the ability of spaceflight to elicit changes in microbial gene expression, including alterations in virulence parameters, is essential in determining microbial risks and identifying options for reducing those risks to crewmembers during long duration International Space Station (ISS), Lunar, and Martian missions. Proper assessment of the microbial risk optimizes the costs associated with mission operation and ensures appropriate health care for the crew. A pre-flight Experiment Verification Test (EVT) was completed 6/29/06 at KSC Space Life Sciences Lab. Post flight the Microbe payload was returned to the KSC Space Life Sciences laboratory, where virulence samples were prepared and used for virulence studies. Microarray samples were taken to PI and CoI laboratories for analyses. The quality of samples received was exceptional and ideally suited for the research in gene expression profiling using microarrays, virulence studies in animals, and morphology analysis via SEM. Results from Microbe are still being analyzed; current data indicates: increased microbial growth in-flight, alterations in microbial gene expression in-flight, alternations in microbial morphology in-flight, alterations in virulence resulting from the flight, changes in virulence related to media composition. The first manuscript form this work is in preparation with plans for submission in May-June 2007.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Johanson K, Allen PL, Gonzales-Villalobos RA, Nesbit J, Nickerson CA, Höner zu Bentrup K, Wilson JQ, Ramamurthy R, D'Elia R, Muse KE, Hammond J, Freeman J, Stodieck LS, Hammond TG. "Haploid deletion strains of Saccharomyces cerevisiae that determine survival during space flight." Acta Astronautica 2007 Feb-Apr;60(4-7):460-71. http://dx.doi.org/10.1016/j.actaastro.2006.09.011 , Feb-2007

Articles in Peer-reviewed Journals	Nauman EA, Ott CM, Sander E, Tucker DL, Pierson D, Wilson JW, Nickerson CA. "Novel quantitative biosystem for modeling physiological fluid shear stress on cells." Appl Environ Microbiol. 2007 Feb;73(3):699-705. Epub 2006 Dec 1. PMID: 17142365 , Feb-2007